

|                                                                                   |                                                                                               |                   |                              |  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|-------------------|------------------------------|--|
|  | Case Name: C2019-21                                                                           | Sector            |                              |  |
|                                                                                   | OR-AS<br>Operations Research -<br>Applications and Solutions<br>www.or-as.be<br>info@or-as.be | Baseline schedule | Schedule with resources      |  |
|                                                                                   |                                                                                               |                   | Schedule with costs          |  |
|                                                                                   |                                                                                               | Risk analysis     | Random Simulation            |  |
| Submitted by                                                                      | Mayté & Thymo                                                                                 |                   | One of nine std. scenarios   |  |
| Date                                                                              | 2019                                                                                          | Project Control   | User defined distributions   |  |
| File Name                                                                         | C2019-21                                                                                      |                   | Automatic tracking           |  |
|                                                                                   |                                                                                               |                   | Tracking based on user input |  |

## 1. Project description

Project Authenticity

The project consists of *activity, resource and cost* data that were obtained directly from the actual project owner.

## 2. Project properties

### 2.1. Baseline schedule

| General                    |                 |
|----------------------------|-----------------|
| # Activities               | 31              |
| Planned Duration (PD)      | 671 days*       |
| Budget at Completion (BAC) | € 13.812.919,00 |
| Renewable Resources        | 3               |
| Consumable Resources       | -               |

| Network Topology           |     |
|----------------------------|-----|
| Serial/Parallel (SP)       | 40% |
| Activity Distribution (AD) | 59% |
| Length of Arcs (LA)        | 92% |
| Topological Float (TF)     | 4%  |

\* Standard eight-hour working days

### 2.2. Risk analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

|          | Cost sensitivity |             |          |
|----------|------------------|-------------|----------|
|          | avg (%)          | std dev (%) | skew (-) |
| CRI-r    | 13,90            | 15,62       | 2,4      |
| CRI-rho  | 13,91            | 14,93       | 2,5      |
| CRI-thau | 10,58            | 10,77       | 2,3      |

|          | Time sensitivity |             |          |
|----------|------------------|-------------|----------|
|          | avg (%)          | std dev (%) | skew (-) |
| CI       | 38,0             | 44,5        | 0,5      |
| SI       | 35,6             | 36,0        | 0,6      |
| SSI      | 9,7              | 15,4        | 1,7      |
| CRI-r    | 12,4             | 14,5        | 2,2      |
| CRI-rho  | 12,6             | 14,1        | 2,3      |
| CRI-thau | 9,8              | 10,3        | 2,0      |

|          | Resource sensitivity |             |          |
|----------|----------------------|-------------|----------|
|          | avg (%)              | std dev (%) | skew (-) |
| CRI-r    | 0,16                 | 0,20        | 3,9      |
| CRI-rho  | 0,16                 | 0,20        | 3,9      |
| CRI-thau | 0,16                 | 0,20        | 3,9      |